

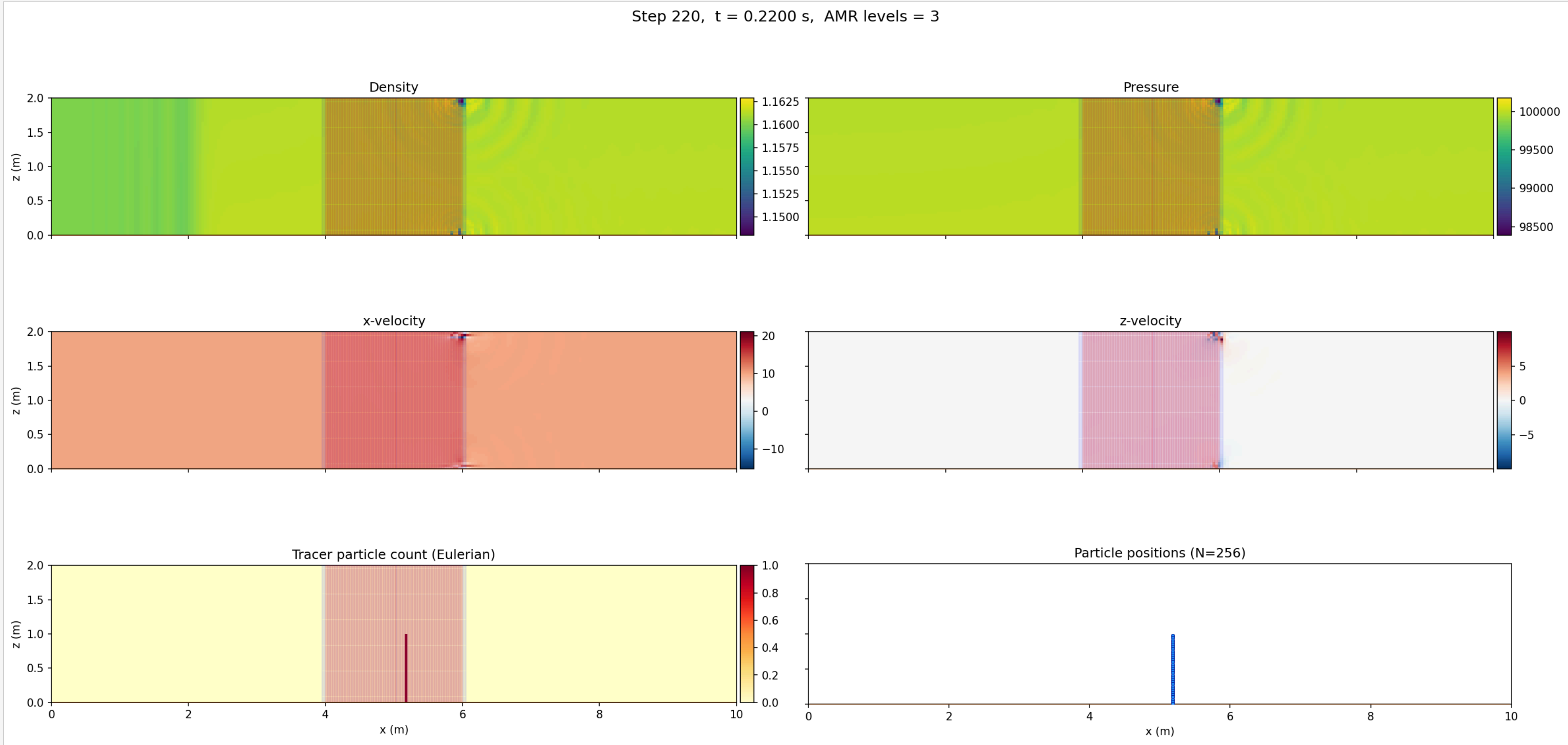
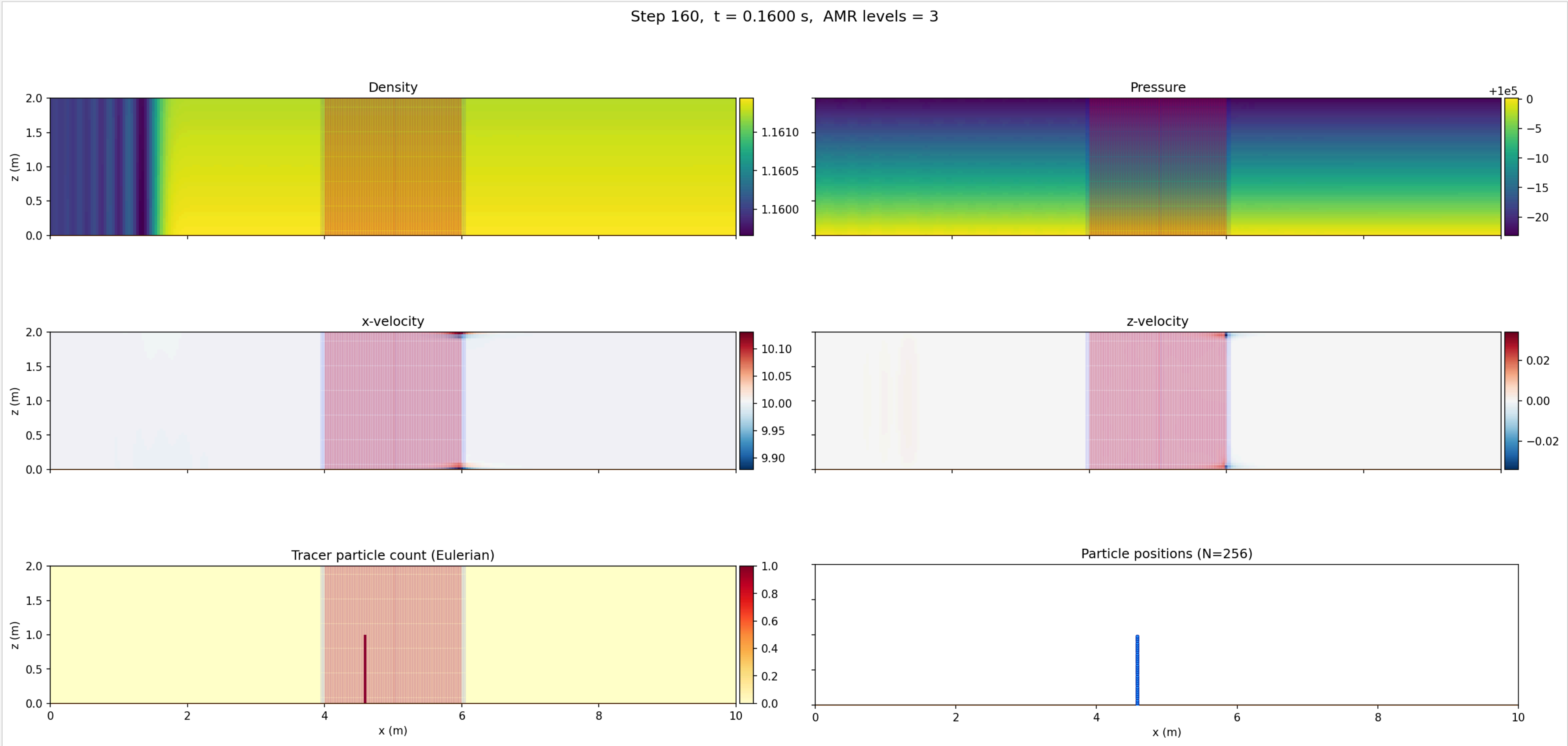
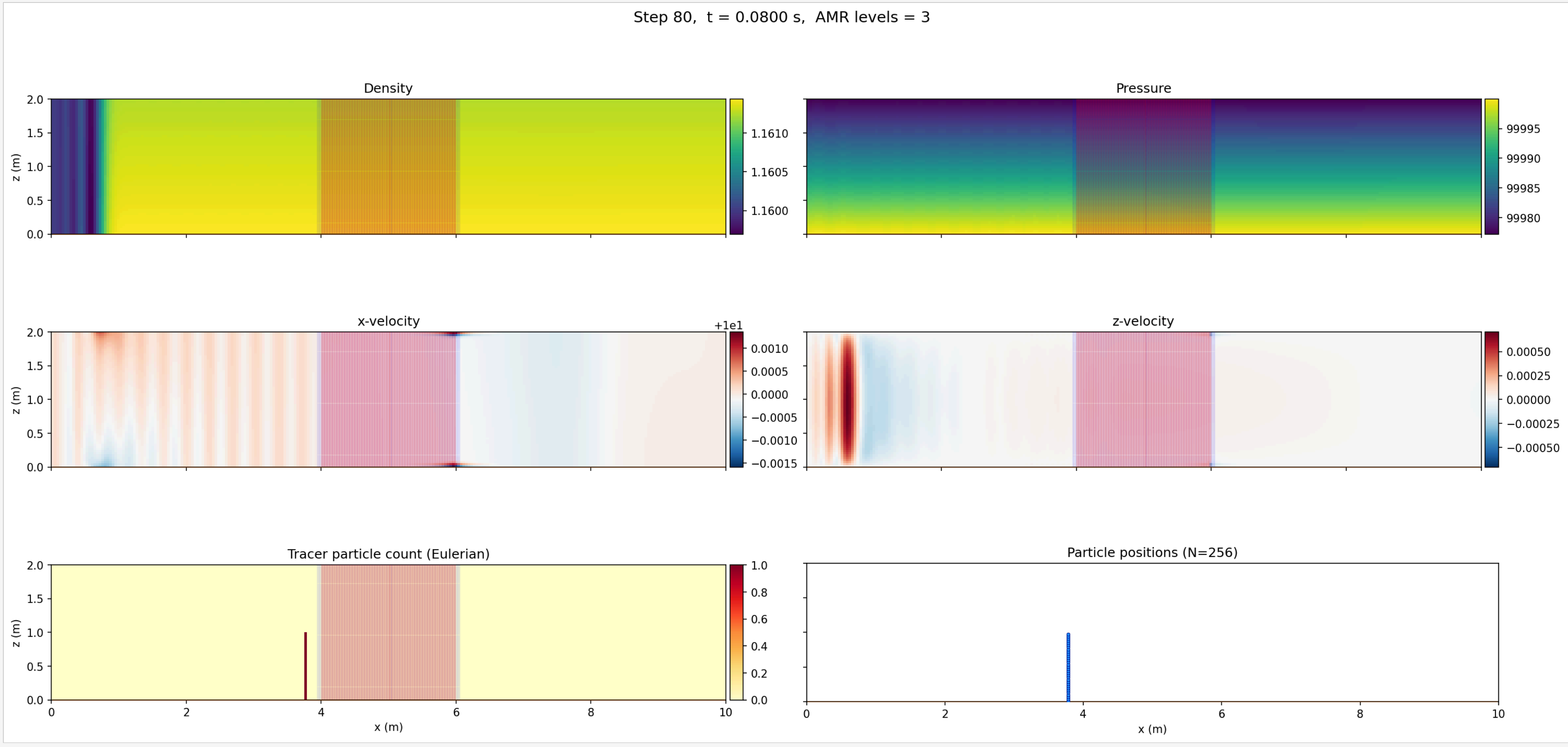
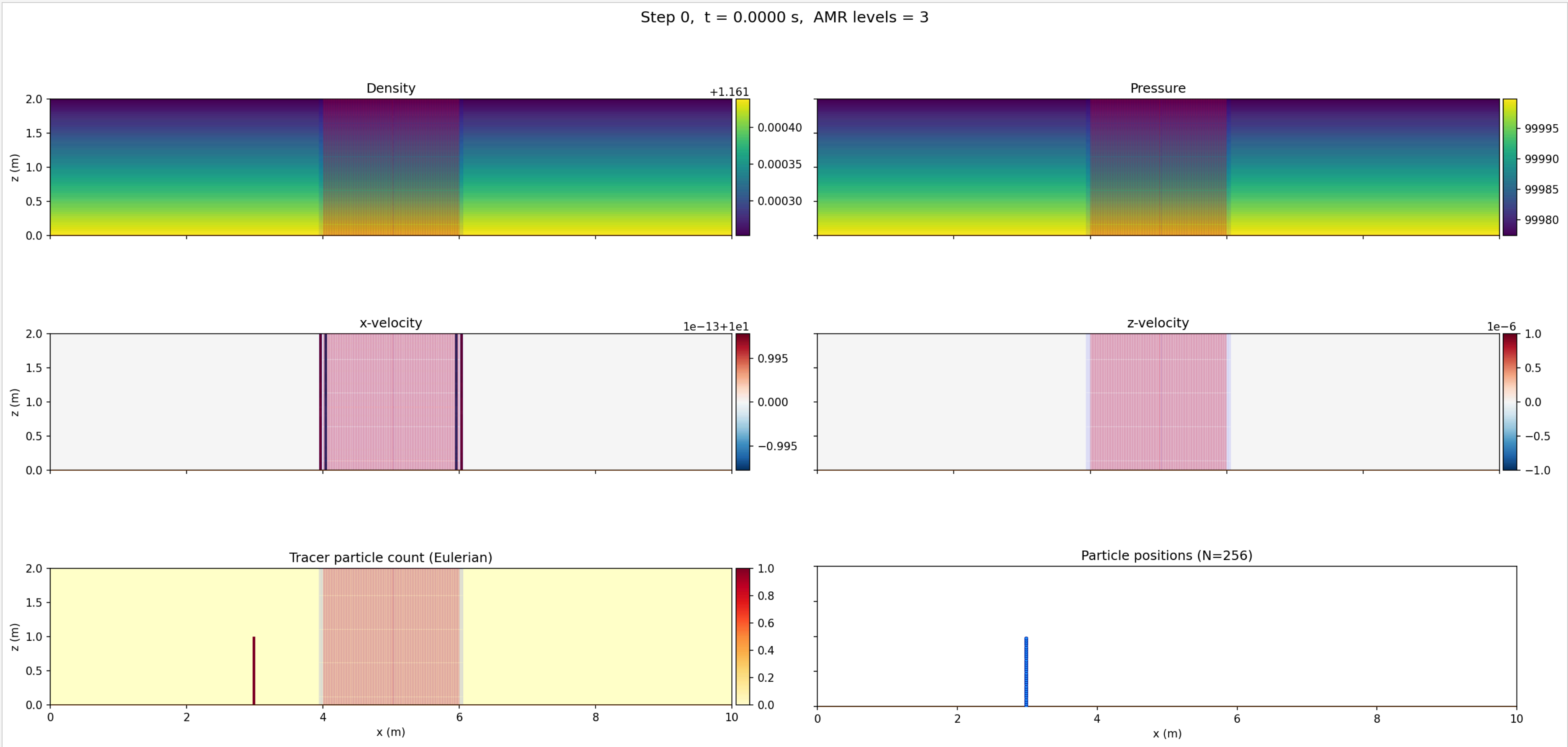
Two-Way Coupling: Failed Cases (2026-04-08)

Branch: dg/particles_w_AMR — 7 of 14 AMR cases failed
All failures are flow instabilities at coarse-fine boundaries.

1. Flat terrain, AMR2, box_fullz

Input: [inputs_over_flat_AMR2_box_fullz](#)
FPE (floating-point exception) at coarse step 237 during L2 time integration. Static box covers full z-extent with 2 refined levels (ratio 2,1,2 each). Flow develops spurious oscillations at the L2 coarse-fine boundary.

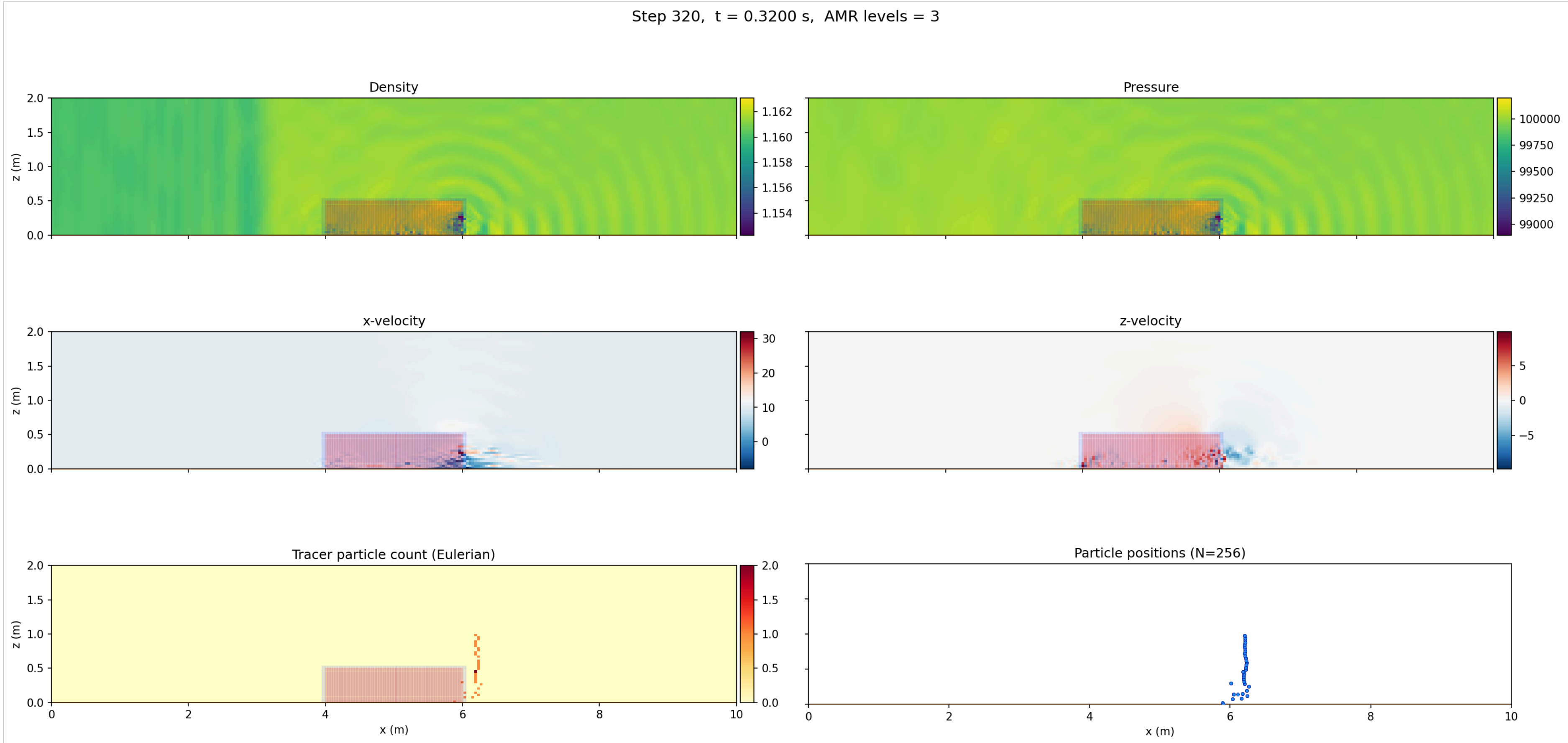
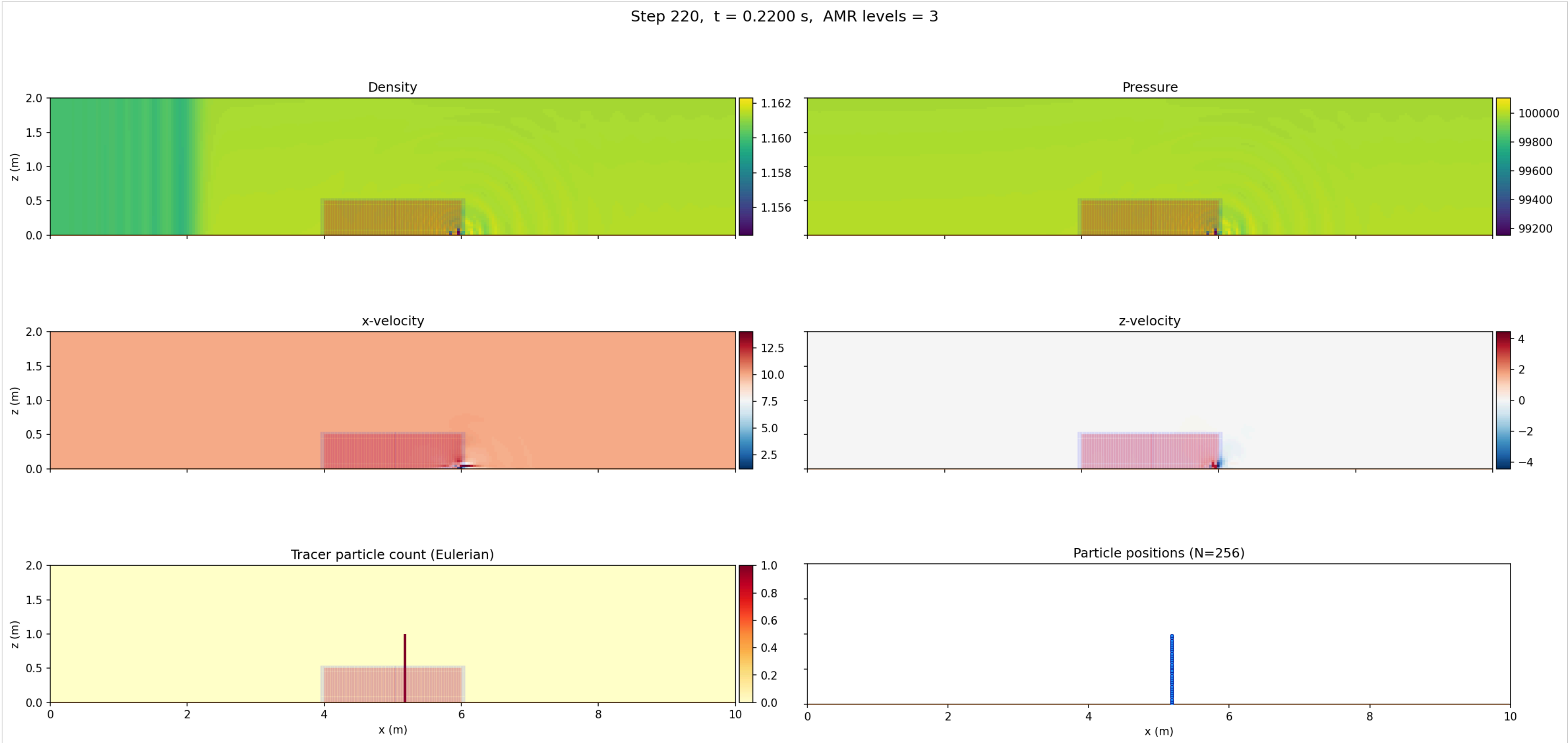
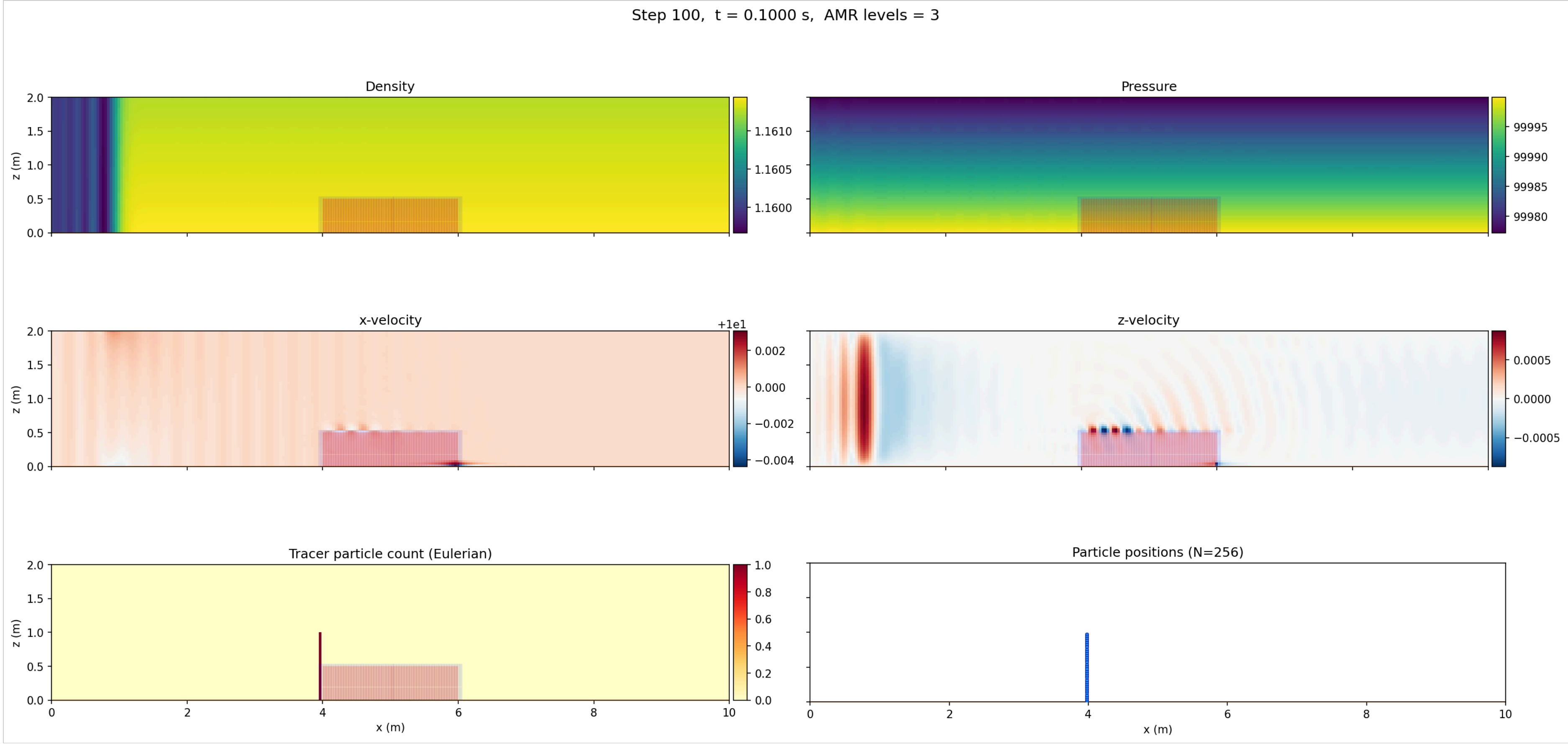
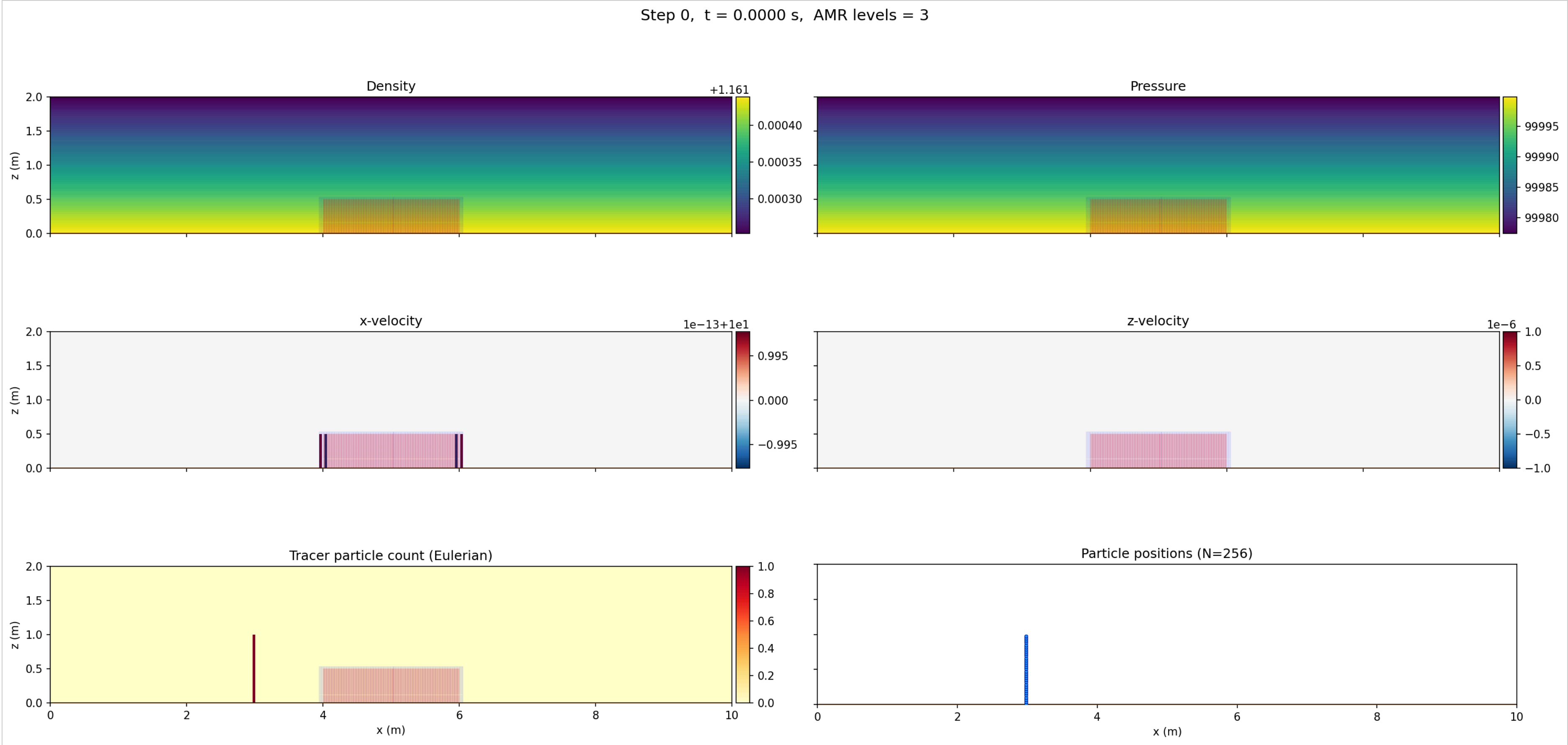
Backtrace: signal caught in [powf64](#) (libm) called from [ERF_TI_utils.H:41](#) inside the slow RHS lambda of [advance_dycore](#) → [ERF_TI_slow_rhs_pre.H:26](#) → [ERF_MRI.H:277](#) (MRI split integrator). The FPE occurs when NaN/Inf values from the C-F boundary reach the exponentiation in the RHS computation.



2. Flat terrain, AMR2, box_partialz

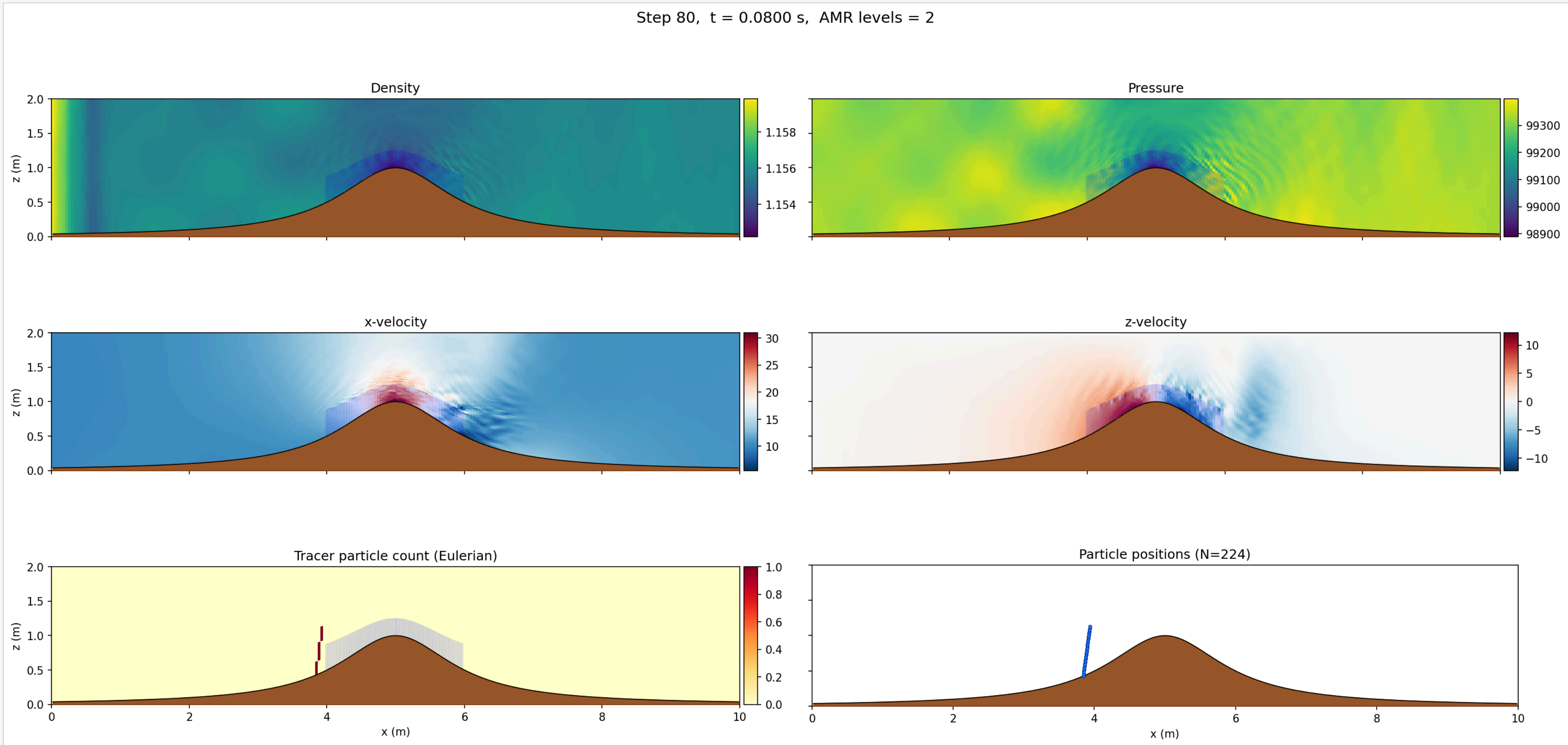
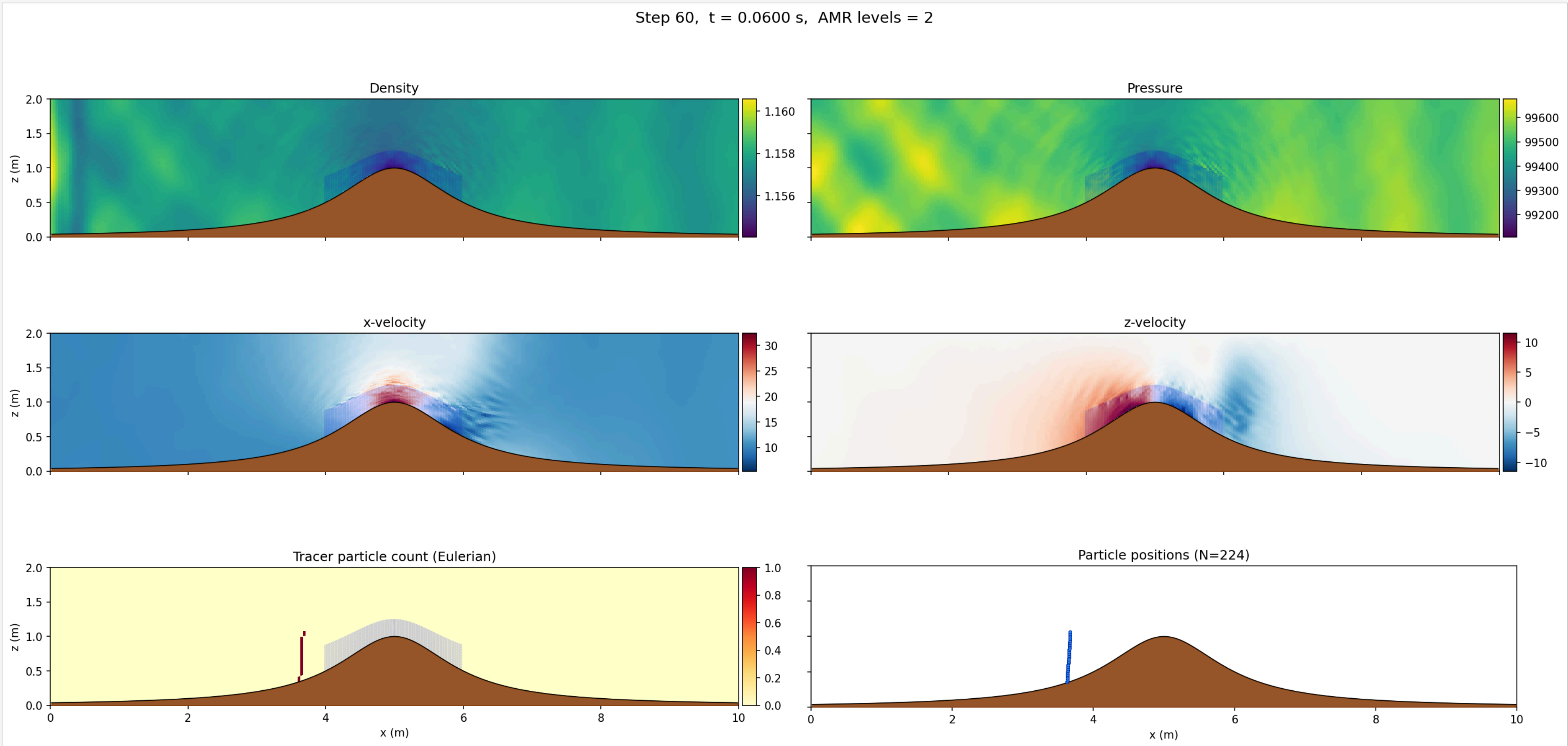
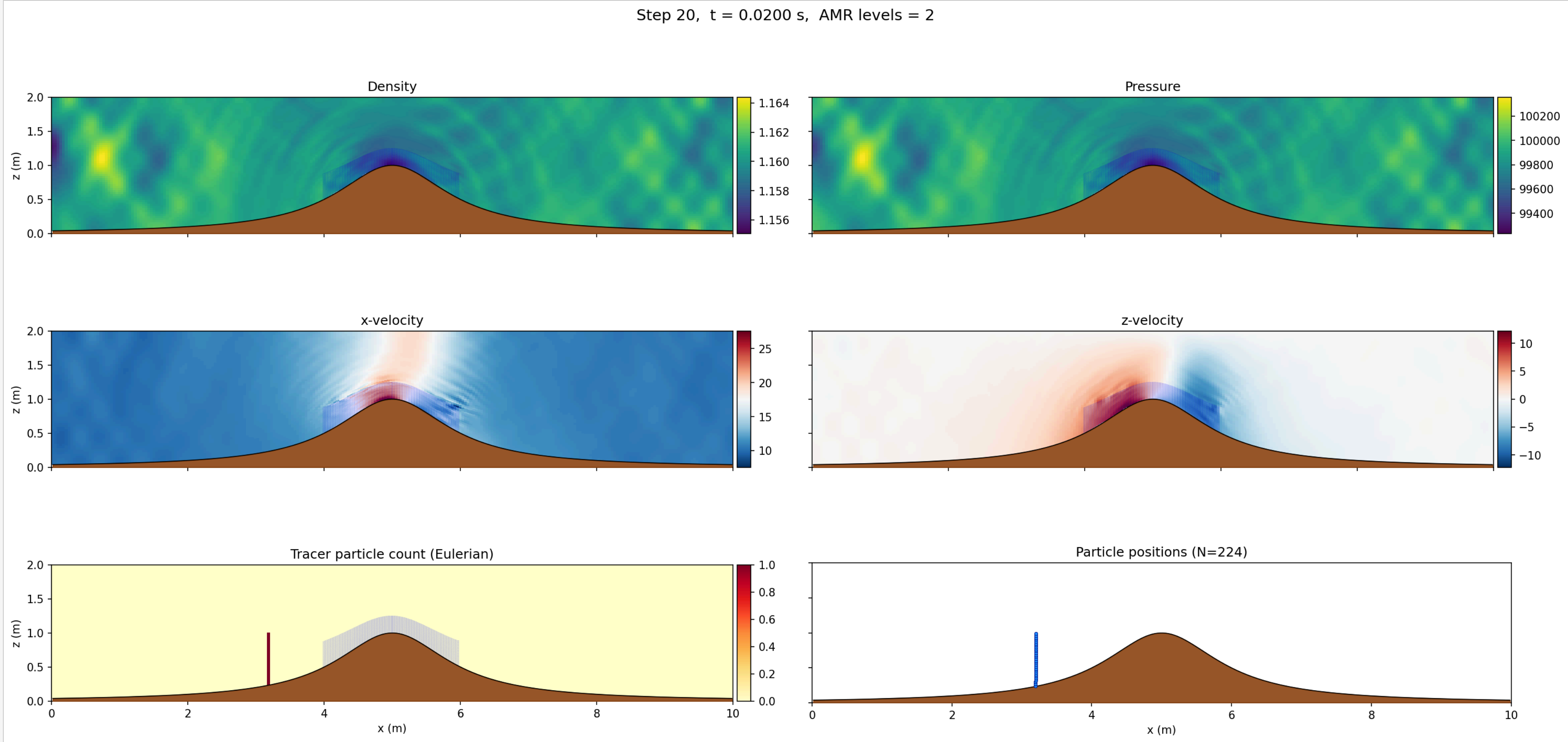
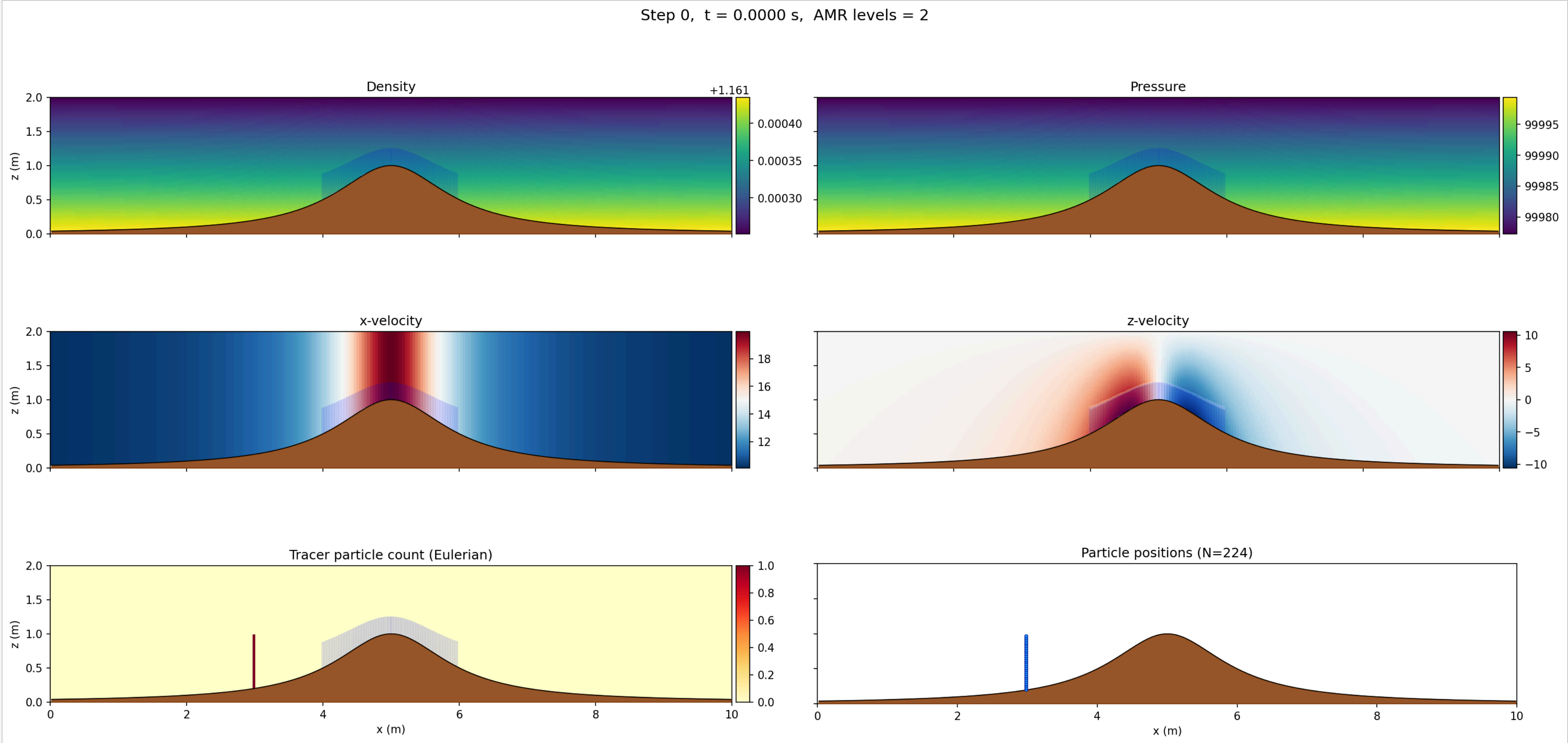
Input: [inputs_over_flat_AMR2_box_partialz](#)
Negative density at coarse step 332 on L2. Static box covers partial z-extent with 2 refined levels. Instability grows at the L2 boundary until density goes negative.

Backtrace: [amrex::Abort](#) called from [ERF::check_for_negative_theta](#) ([ERF.cpp:3237](#)) after [ERF::Advance](#) ([ERF_Advance.cpp:269](#)). Message: "Bad rho in check_for_negative_theta" — negative density at cell (613, 0, 37) with rho = -0.469.



3. Terrain (WoA), AMR1, box_partialz

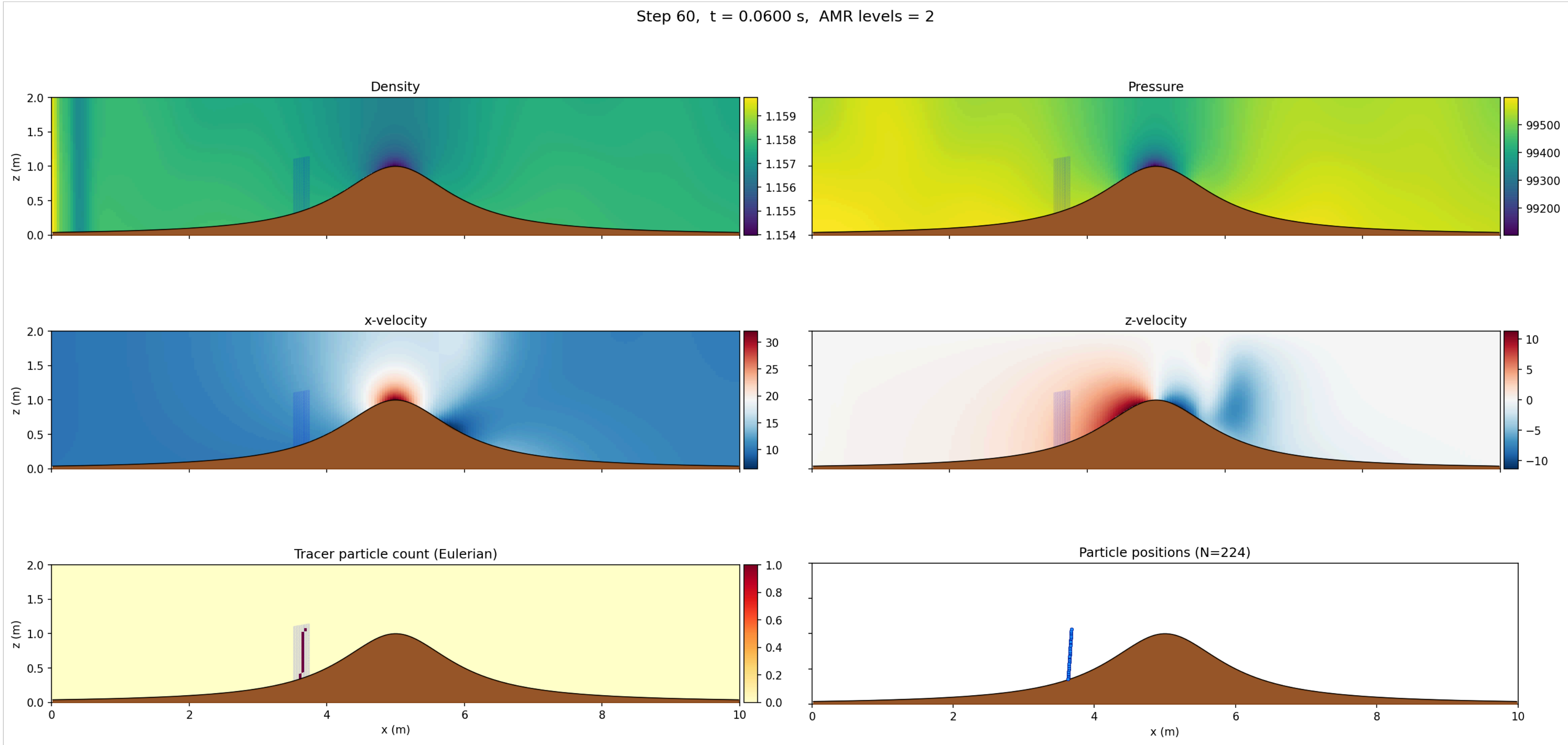
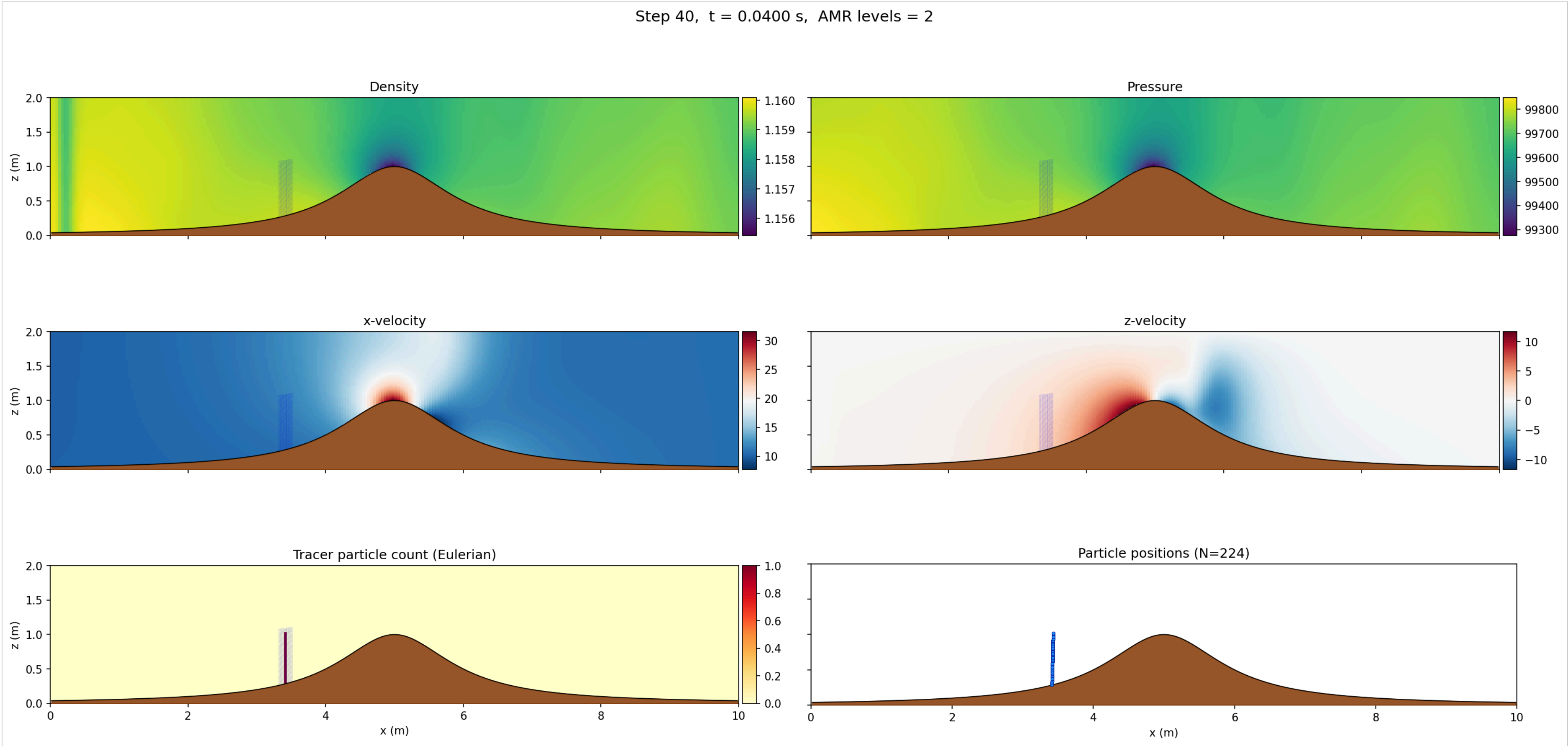
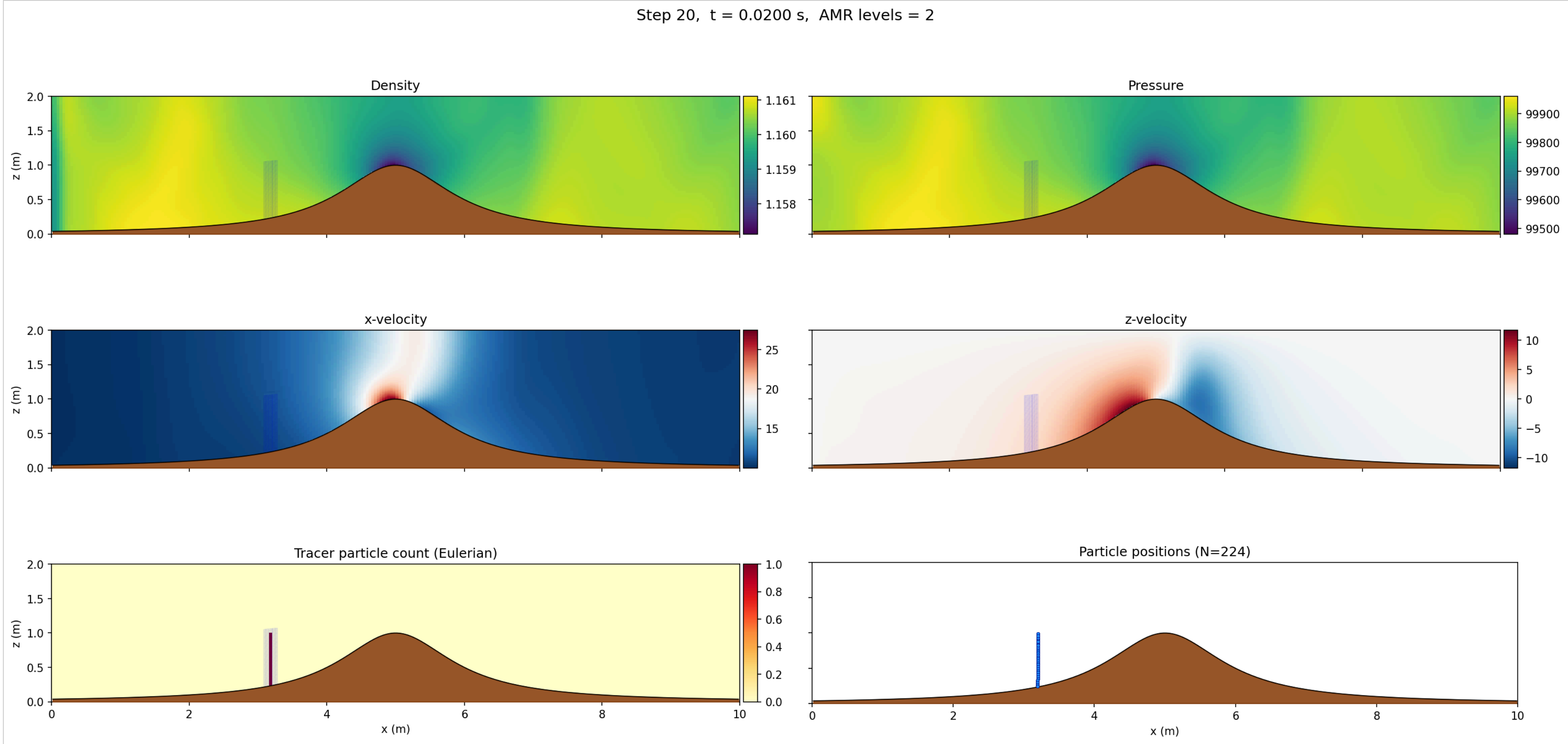
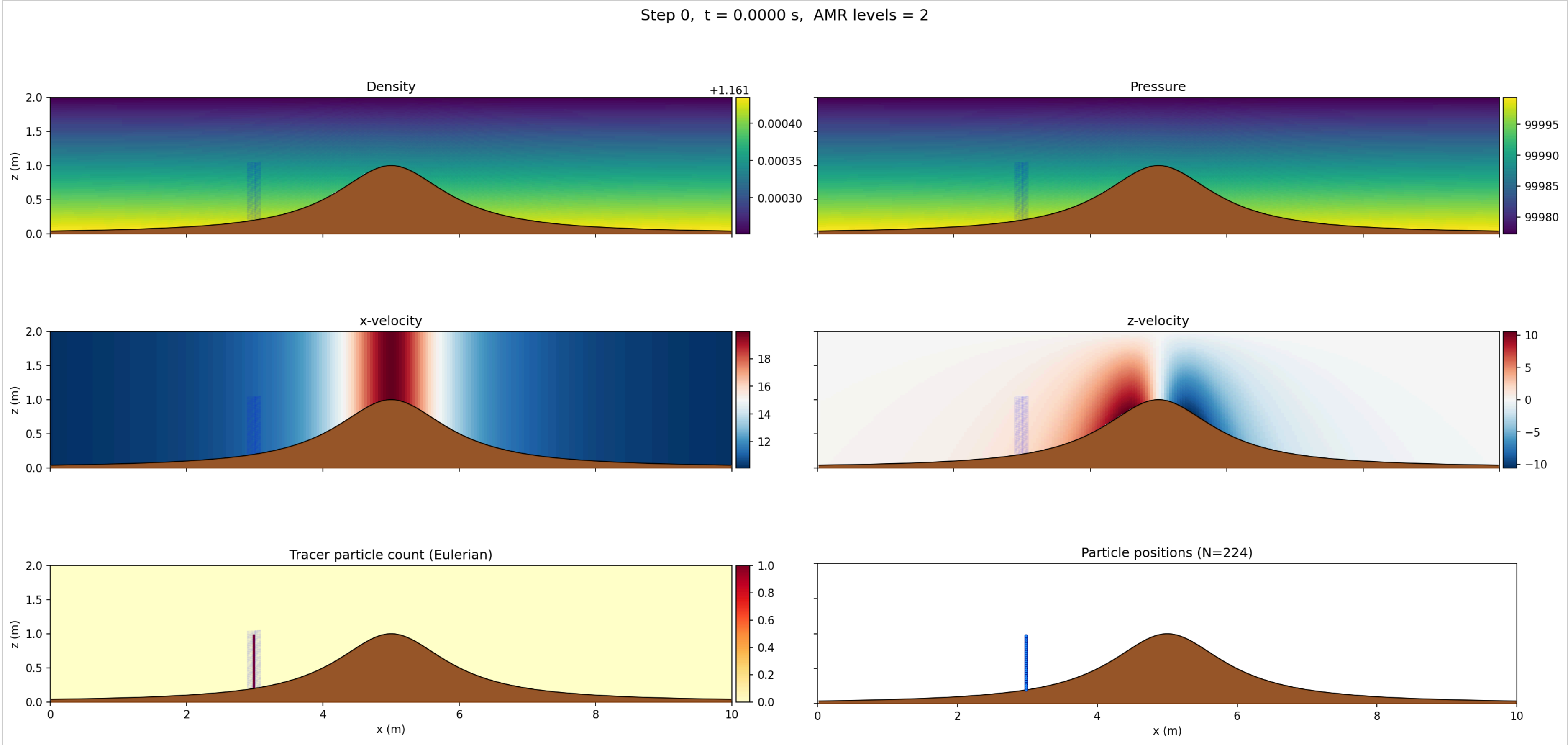
Input: `inputs_over_hill_AMR1_box_partialz`
Heap corruption at coarse step 100. Terrain-following coordinates with partial-z refined box. Index-space interpolation at the top of the refined region produces incorrect values because it assumes uniform z-spacing.
Backtrace: "free(): invalid next size (fast)" triggered during `ERF::Write3DPlotFile ( ERF_Plotfile.cpp:1473 )` → `__libc_free`. The heap corruption originates earlier during time integration; the plotfile writer merely triggers it when freeing memory. The flow had already been corrupted by bad C-F interpolation.



4. Terrain (WoA), AMR1, particlecount

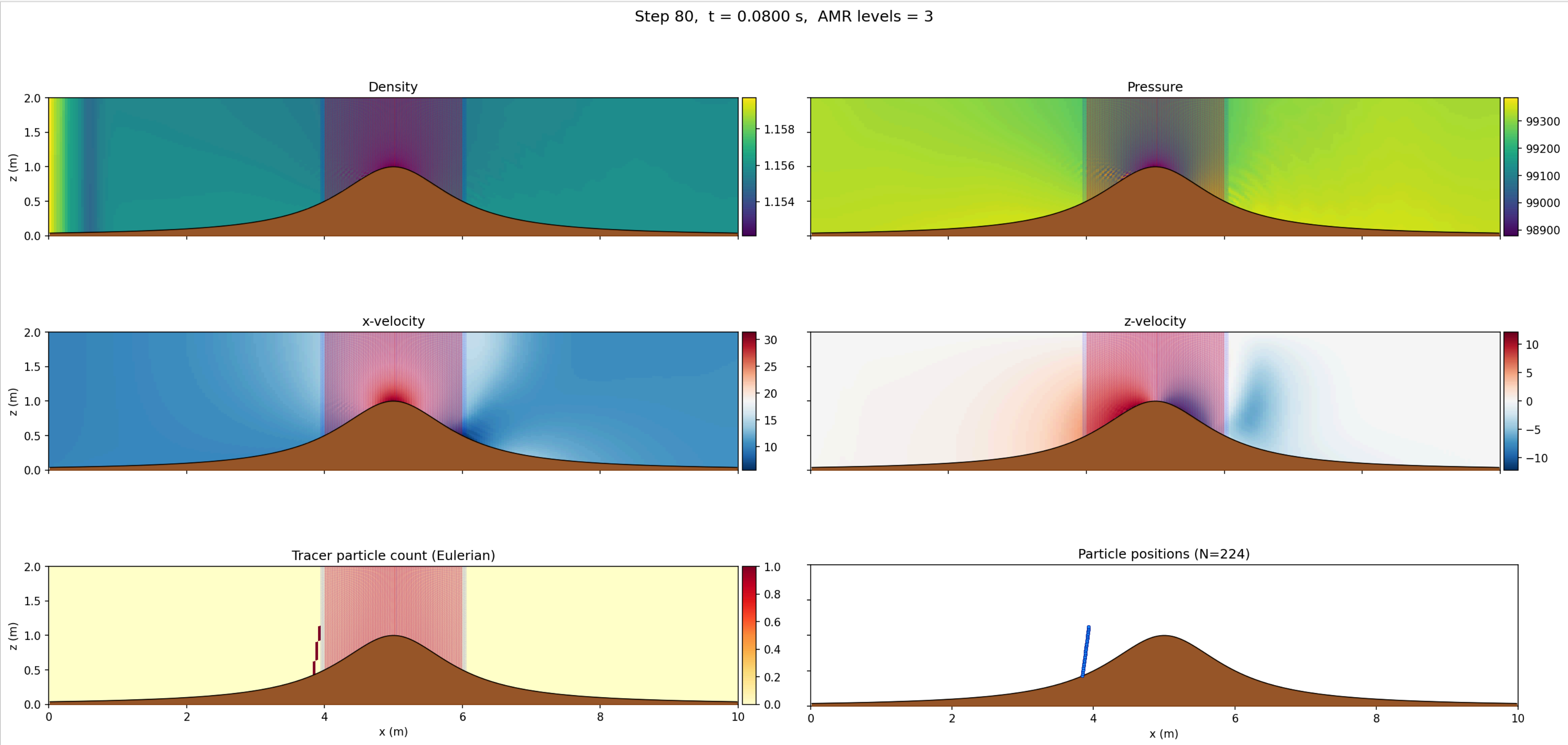
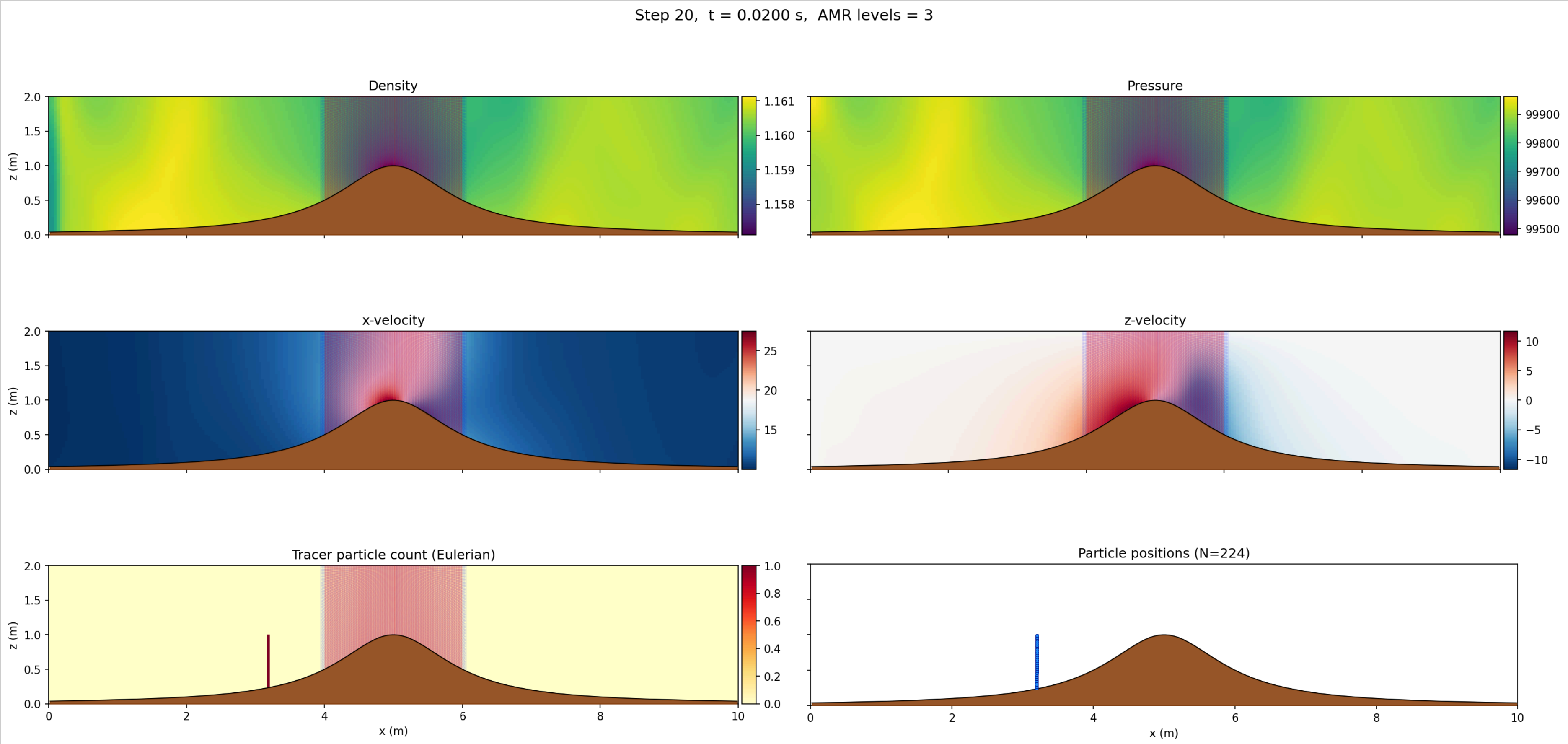
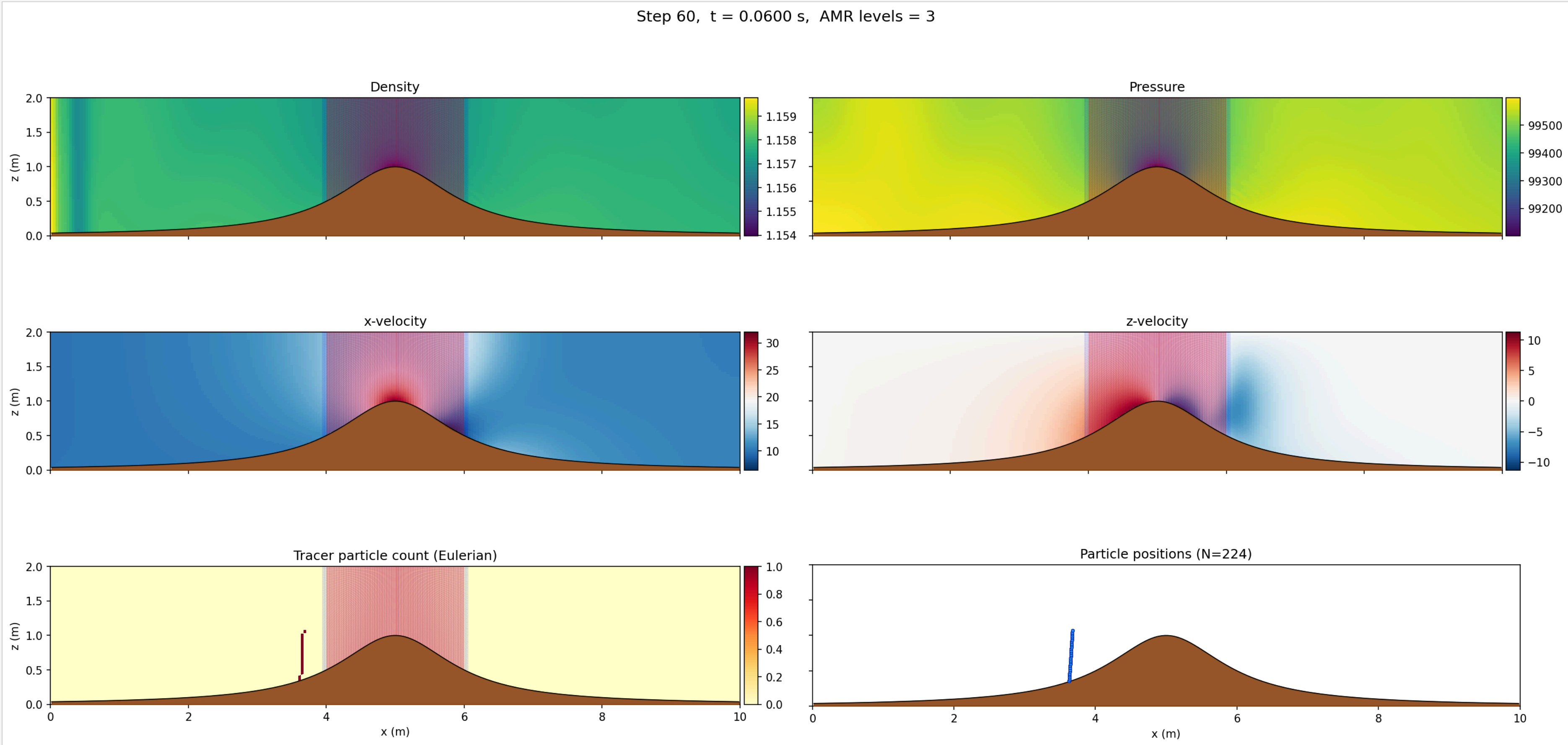
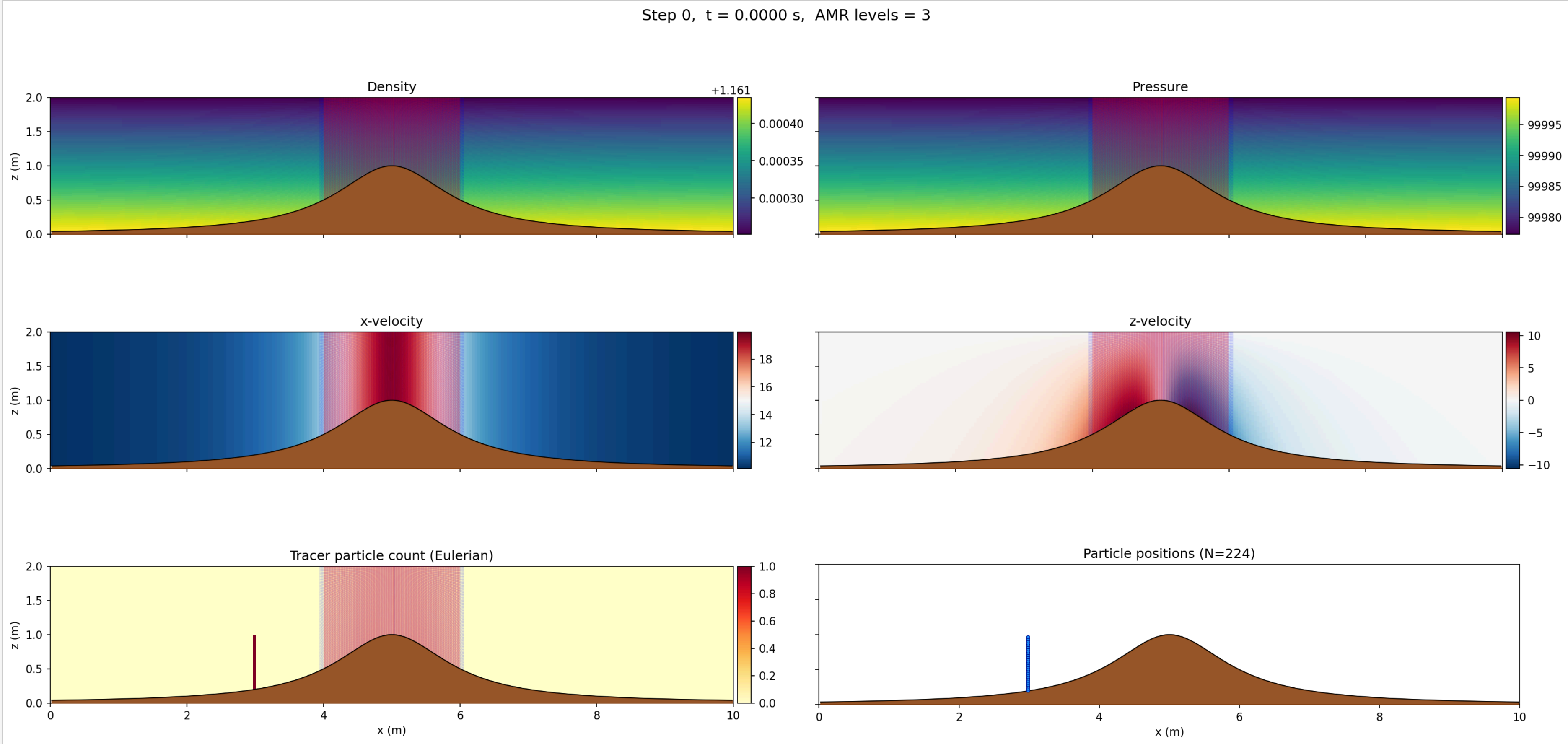
Input: [inputs_over_hill_AMR1_particlecount](#)
Segfault at coarse step 61 during regrid. Dynamic particle-count tagging with terrain-following coordinates. The refined region moves with the particles, and the coarse-fine interpolation for newly created fine cells uses incorrect z-offsets.

Backtrace: segfault in [FabArrayBase::getCPC](#) ([AMReX_FabArrayBase.cpp:593](#)) during [ParallelAdd](#) inside [IncrementWithTotal](#), called from [ERF::ErrorEst](#) ([ERF_Tagging.cpp:301](#)) → [AmrMesh::MakeNewGrids](#) → [AmrCore::regrid](#). Crash occurs during particle count deposition for the tagging criterion.



5. Terrain (WoA), AMR2, box_fullz

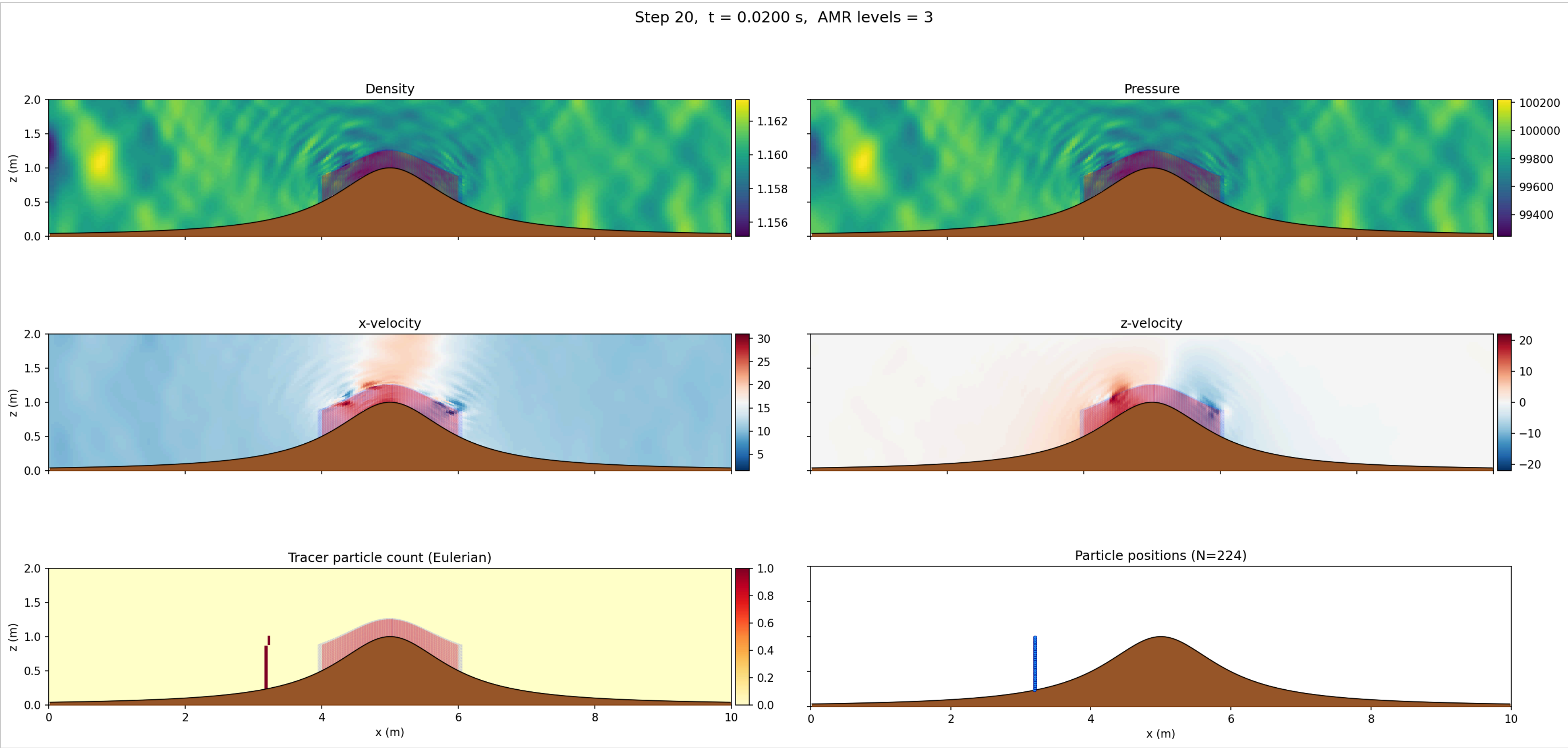
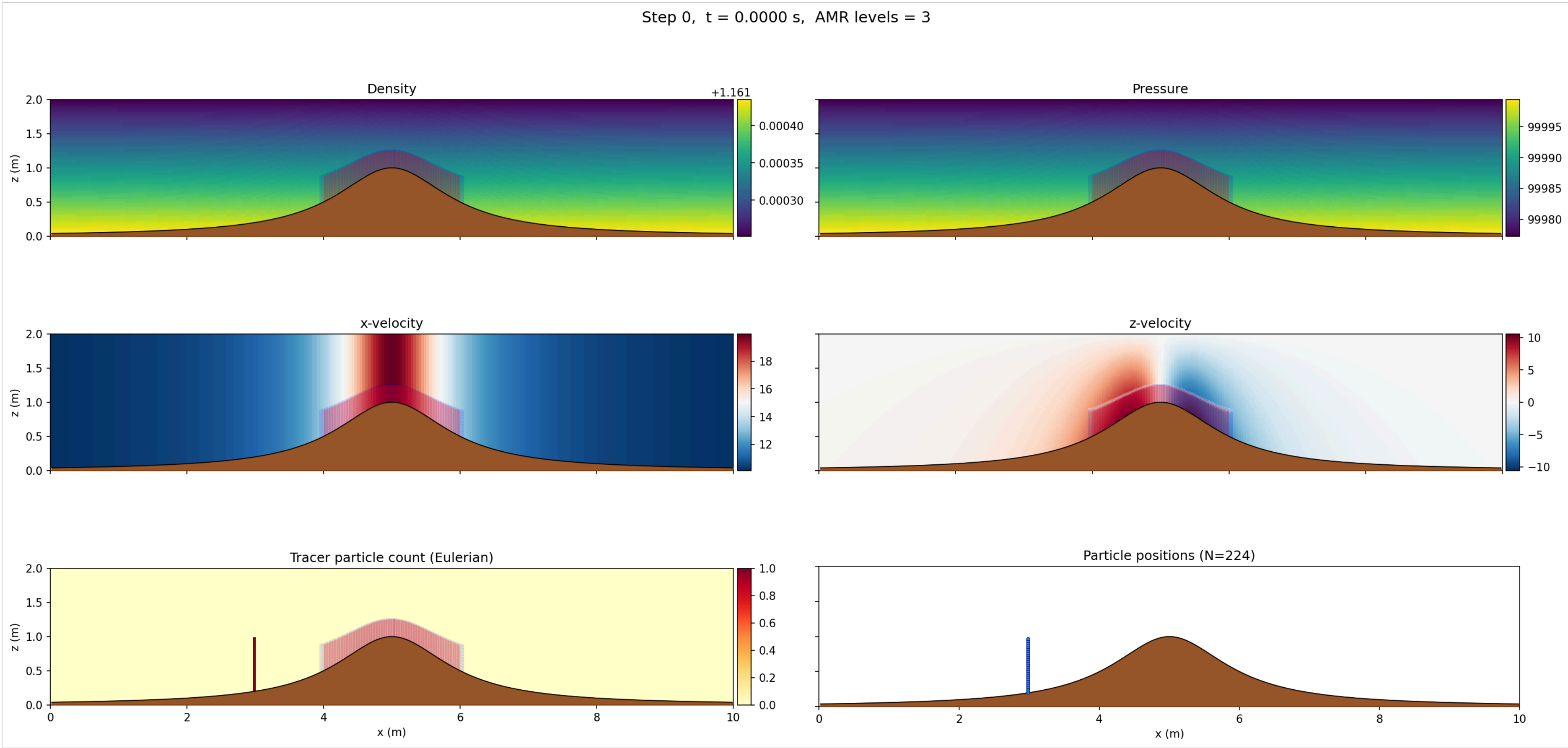
Input: [inputs_over_hill_AMR2_box_fullz](#)
FPE at coarse step 85 during L2 time integration. Two refined levels with terrain-following coordinates. Even with full-z box coverage, 2 levels of refinement amplify the interpolation error.
Backtrace: same as case 1 — signal in [powf64](#) (libm) from [ERF_TI_utils.H:41](#) → [ERF_TI_slow_rhs_pre.H:26](#) → [ERF_MRI.H:277](#) → [ERF_AdvanceDycore.cpp:356](#) → [ERF_Advance.cpp:249](#) . NaN/Inf from incorrect terrain C-F interpolation propagates into the RHS.



6. Terrain (WoA), AMR2, box_partialz

Input: [inputs_over_hill_AMR2_box_partialz](#)
FPE at coarse step 36 during L2 time integration. Earliest failure of all cases. Partial-z box + terrain + 2 refined levels is the most challenging configuration for the interpolation. Only 2 plotfiles were written before the crash.

Backtrace: same as cases 1 and 5 — [powf64](#) → [ERF_TI_utils.H:41](#) → [ERF_TI_slow_rhs_pre.H:26](#) → [ERF_MRI.H:277](#) → [ERF_AdvanceBycore.cpp:356](#). Fails earliest because partial-z + terrain maximizes the z-offset error in the coarse-fine interpolation.



7. Terrain (WoA), AMR2, particlecount

Input: [inputs_over_hill_AMR2_particlecount](#)
Heap corruption at coarse step 101 during regrid. Dynamic tagging + terrain + 2 refined levels. Crashes during level remeshing when coarse-fine interpolation fills newly created fine cells with bad data.

Backtrace: "munmap_chunk(): invalid pointer" in [__libc_free](#) called from [FabArrayBase::flushFB](#) ([AMReX_FabArrayBase.cpp:1272](#)) during [ParticleContainerBase::RedefineDummyMF](#) ([AMReX_ParticleContainerBase.cpp:68](#)) → [RedistributeCPU](#) → [ERFPC::FixIndexAMR](#) ([ERFPCUtils.cpp:174](#)). Heap was corrupted by prior bad data from terrain C-F interpolation; the particle redistribute after regrid triggers the crash.

